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Project Proposal

Physics-Informed Synthesis of 3D Hand Trajectories for Gestural Musical Control in Rope Flow

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Motivation and research question

Traditional digital interfaces for music often rely on discrete triggers or simplified planar movements (e.g., pressing a button). However, athletic activities like rope flow (i.e., a cyclic athletic movement where the performer rotates a weighted rope in periodic patterns) involve complex, high-dimensional coiling gestures that represent a rich source of expressive data. While the rhythm of these movements is a primary component, it is not the only dimension; the spatial topology, velocity profiles, and phase relationships between the performer's hands offer a diversified space for musical synthesis.

Therefore, this project aims to build a model that transforms human movement (rope flow) into sound. It specifically answers the question of **how to accurately reconstruct 3D hand movement using a physics-informed data-driven model and enable real-time musical sonification?** The goal is to develop a robust physics-informed model that classifies rope flow patterns and maps their oscillation dynamics (e.g., frequency, amplitude, and phase) to musical parameters like BPM, volume, and pitch. Consequently, this work moves performers from passive listeners to active composers who can transform their movement kinematics into real-time musical expression.

Background and existing literature

Modern dynamics discovery relies on frameworks like Physics-Informed Neural Networks (PINNs) (Raissi et al., 2019) and SINDy (Brunton et al., 2016) to embed physical laws into data-driven models. Recent advancements target inertial drift in periodic motion: Sahoo and Klein (2025) utilize overlapping data windows in MoRPI-PINN to constrain equations of motion, while Xue et al. (2025) mitigate drift by fusing IMU data with distance constraints via state-space models. To ensure biomechanical validity, Bian et al. (2026) embed orientation-based constraints directly into neural architectures, providing a template for the physically consistent trajectory reconstruction required for rope flow.

In the context of gesture-to-sound mapping, the field has evolved from linear SVD-based methods (Bevilacqua et al., 2005) and real-time HMM gesture followers (Bevilacqua et al., 2010) to deep learning approaches. While Caramiaux and Tanaka (2013) established the groundwork for ML in music, newer research by Erdem et al. (2020) and Zhao et al. (2018) utilizes LSTMs and ResNets to model complex motion-sound energy relationships. However, these "black-box" models often lack the interpretable control signals necessary for nuanced performance. Mitcheltree et al. (2025) address this by using differentiable modulators to create human-readable mapping parameters. **Therefore, this project contributes to this lineage by first applying PINN architectures to the kinematic and periodic constraints of rope flow to enable physically grounded trajectory reconstruction, as well as by using classification to map sound alterations to motion.**

Data description and collection

The project utilizes a pre-collected experimental dataset consisting of 14 sessions of rope flow performance (30 seconds to 1 minute each), with plans to expand to 30 sessions. The setup

is already in place and the data is captured using two M5Stick C Plus 1.1 microcontrollers – one at each hand – recording 6-axis IMU (Inertial Measurement Unit) time series data. All video recordings and IMU data for each session are collected and saved using a pre-designed application. The IMU data includes linear acceleration: $\mathbf{a(t)} = [a_x, a_y, a_z]$ and angular velocity: $\boldsymbol{w}\mathbf{(t)} = [w_x, w_y, w_z]$ for each hand and their associated timestamps. A sample data subset extracted from the csv file for one rope session is provided in Appendix C of this proposal.

Methodology

The primary challenge to address in this project is the IMU integration drift. While sensors like the M5Stick provide high-resolution acceleration data, the mathematical process of double-integrating these signals to find spatial position is prone to exponential error accumulation. Therefore, we plan to apply a Madgwick filter for data pre-processing to fuse the accelerometer and gyroscope data and establish a stable orientation estimate.

Subsequently, a PINN-based integrator is used to estimate 3D trajectories and reconstruct the hand movement p(t). To prevent drift, we embed kinematic and periodic constraints as regularization terms in the loss function. By minimizing a composite loss function $L = L_{data} + \lambda L_{physics}$, where $L_{physics}$ penalizes any position that contradicts the inductive biases (listed at the end of this section), we can generate a spatial mapping. In this step, we apply cross-validation using the initial set of recordings (80% of total data).

Then, for the reconstructed p(t), we extract features for pattern recognition such as the topology of the trajectory (e.g., infinity shape), spectral features using FFT-based dominant frequencies, or kinematic features like peak velocity and acceleration. A classification model, potentially Random Forest or 1D-CNN, is trained on the reconstructed trajectories to identify specific patterns.

The final stage involves a mapping function from movement space *M* to musical space *S*. Classified patterns are mapped to trigger specific MIDI instruments, for example the frequency of the IMU peaks can dictate the tempo of the audio output.

The methodology described above incorporates inductive biases that restrict the hypothesis space, improve generalization, and mimic the underlying physics. The spatial constraint limits the rope kinematics by rope length and human joint limits using a log-barrier $L_{boundary} = \text{ReLU}(\| p(t) - p_{shoulder} \| - R_{arm})^2$. In addition, given the periodicity of rope flow, the integral of velocity over a full cycle T must satisfy $\int_t^{t+T} v(\tau)\, d\tau \approx 0$ to prevent the hand from drifting infinitely into space by enforcing periodic boundary conditions on the position p(t) = p(t + T). Moreover, human movement naturally minimizes jerk (the third time derivative of position), which can be mathematically reflected in the loss function as $\int \| \frac{d^3p}{dt^3} \|^2 \, dt$ to naturally suppress the "jitter" caused by integration drift.

Evaluation strategy

The performance of the model will be evaluated using two primary metrics. For the classification of patterns, we will use an F1-score to measure how accurately the model identifies

the "Flow Pattern". For the reconstruction of dynamics, we will calculate the Root Mean Square Error (RMSE) between the drift magnitude of the PINN-reconstructed trajectory over a 60-second window compared to a double-integration approach. We will ensure that enough rope flow sessions are used as a test set to report the robustness of the model. For the sonification part, we will investigate the time-to-peak error between the IMU data peaks at t_1 and the synthesized audio peaks at t_2 and employ the metric $|t_1 - t_2|$ to determine the phase alignment of the motion to music mapping.

Process and timeline

The project is structured over the coming nine-week timeline such that:

- Week 1–2: Signal processing and denoising of sensor data.
- Weeks 3–4: Development of the PINN integrator with the implementation of inductive biases based on the initial data of the 14 sessions. Expansion of data collected (up to 30 sessions).
- Weeks 5–6: Feature engineering. Pattern classifier training and sonification mapping.
- Week 7: MIDI integration and latency/jitter optimization
- Week 8: Testing (different users) and evaluation of a data subset. Model refinement according to testing results.
- Week 9: Finalization of deliverables and report writing.

Project deliverables

Project deliverables include a progress report by April 15 to detail out the success of the trajectory reconstruction, in addition to a final presentation on April 29 – May 1, and a demonstration of the "Motion-to-Music" system. The final report on May 5 will constitute a comprehensive paper laying out the end-to-end process and results of the data-driven discovery of rope dynamics and movement sonification.

References

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Appendices

Appendix A: LLM conversation

Two LLM conversations were conducted independently. One using Claude (attached as a separate pdf) and another one using Gemini 3 provided on the link hereby: https://gemini.google.com/share/677be117b4d1

Appendix B: Brief description of manuscript preparation methodology

The development of this proposal was supported by an iterative conversation with LLM. The manuscript was prepared as follows:

- Preliminary scoping

The process began by providing the LLM with the initial project vision. This allowed us to identify generic suggestions (like basic Madgwick filtering) that needed to be challenged or expanded upon to achieve research-grade depth.

- Critical questioning and feedback

We explicitly prompted LLM to act as a skeptical peer reviewer. We pushed it to find flaws in the original methodology, specifically regarding double integration drift and hard geometric constraints. This stage was critical in shifting the project from a naive mathematical approach to a more robust PINN framework.

- Literature deep-dive & gap analysis

Instead of accepting the LLMs first list of references (which skewed toward older studies), we challenged the tool to identify recent shifts in the field (2022–2026). The LLM was used to cross-reference technical sensor fusion papers with deep learning and sonification literature to ensure the proposal has both engineering rigor and a sound musical foundation.

- Technical refinement & parameter mapping

The final stage of the conversation focused on "high-resolution" technical details. We iterated the specific components of the loss function, moving beyond simple position tracking. This ensured that conversation moved from high-level ideas to an implementable methodology.

- Reference verification and correction

A dedicated LLM conversation was used to systematically verify each bibliographic reference against web sources. This process identified several errors in the original reference list: Bevilacqua et al. was corrected from 2010 to 2005 with the accurate title and page numbers; Caramiaux & Tanaka was updated with the correct subtitle and pages (513–518); Madgwick et al. was corrected to include "and MARG" in the title; and a fabricated reference (Zhao et al., 2019, "Deep learning for IMU-based human activity recognition") with an invalid DOI was identified and replaced with two verified papers — Erdem et al. (2020) and Zhao et al. (2018) — that accurately support the claims made in the text. The corresponding paragraph in the Background section was rewritten to reflect these corrections.

Appendix C: Example of raw IMU data

timestamp	device	ax	ay	az	gx	gy	gz
1.77E+12	1	0.396729	-0.62305	1.171387	-19.0854	70.487808	-81.7683
1.77E+12	0	-0.52539	-0.58228	1.154541	-18.0488	116.34146	-91.0366
1.77E+12	1	0.389404	-0.57788	1.129395	-32.3171	67.682929	-37.6829
1.77E+12	1	0.469971	-0.4585	1.022461	-57.7439	71.036590	-31.9512
1.77E+12	0	-0.51831	-0.54297	1.141113	-63.7805	89.085365	-80.1829
1.77E+12	1	0.436279	-0.46509	0.99585	-87.1951	78.47561	-43.7195
1.77E+12	0	-0.51123	-0.6355	1.101074	-92.5	65.609756	-84.0854
1.77E+12	0	-0.40845	-0.59839	0.989258	-102.439	46.219512	-87.8049
1.77E+12	1	0.333008	-0.5249	1.020508	-97.1341	67.621955	-32.2561
1.77E+12	0	-0.38867	-0.57544	0.907959	-102.134	29.634147	-84.8171
1.77E+12	1	0.30127	-0.53125	1.038818	-80.8537	56.158538	-3.10976
1.77E+12	1	0.320068	-0.49902	0.986328	-55.6098	66.890243	21.95121955871582
1.77E+12	0	-0.32983	-0.55493	0.798584	-88.1707	5.1219511	-79.6341
1.77E+12	1	0.34082	-0.43311	0.885742	-36.3415	73.048782	37.9878
1.77E+12	1	0.283447	-0.37476	0.798096	-29.6951	65.426834	45.79268264770508
1.77E+12	0	-0.29248	-0.52954	0.682617	-67.9268	-5.85366	-73.1707
1.77E+12	0	-0.26416	-0.49414	0.587891	-53.6585	-8.41463	-70.4268
1.77E+12	0	-0.23584	-0.45825	0.527588	-43.5366	-12.6829	-67.0122
1.77E+12	1	0.251953	-0.37305	0.820313	-30.9146	53.719512	47.9878
1.77E+12	0	-0.28174	-0.45117	0.493896	-32.1341	-10.8537	-53.1707
1.77E+12	1	0.172607	-0.36011	0.785889	-16.1585	45	50.73170852661133
1.77E+12	1	0.081055	-0.33521	0.758789	-1.70732	33.414634	60.121952056884766
1.77E+12	0	-0.2915	-0.49487	0.46875	-27.3171	-5.36585	-42.9878
1.77E+12	1	0.056396	-0.28101	0.784912	1.7682926	22.073171	62.86586
1.77E+12	0	-0.29785	-0.53931	0.462891	-16.9512	1.8902438	-46.0976
1.77E+12	0	-0.31323	-0.54053	0.453369	-5.2439	7.4390244	-45.9146
1.77E+12	1	0.037354	-0.22314	0.804932	2.6829268	12.134146	54.817073822021484
1.77E+12	1	0.00293	-0.18628	0.803955	-0.36585	5.3048782	35.975608825683594
1.77E+12	0	-0.31812	-0.52686	0.423828	1.7682926	10.670731	-44.878
1.77E+12	1	-0.04907	-0.22729	0.843018	-1.34146	-1.40244	23.841463088989258
1.77E+12	0	-0.3269	-0.52319	0.403809	-0.36585	22.012195	-41.1585
1.77E+12	0	-0.32837	-0.51636	0.413086	0.8536585	39.207317	-28.0488
1.77E+12	1	-0.06152	-0.25415	0.865234	3.2926828	-1.64634	18.35365867614746
1.77E+12	0	-0.35132	-0.54785	0.456299	-2.0122	48.292682	-20.6098
1.77E+12	1	-0.10181	-0.26685	0.908936	12.682927	-3.65854	20.670732498168945
1.77E+12	1	-0.08325	-0.27197	0.927979	26.890245	1.9512195	26.03658676147461
1.77E+12	1	-0.11108	-0.27979	0.898438	36.341464	-0.54878	32.256099700927734
1.77E+12	0	-0.38306	-0.62402	0.556641	14.207317	58.963417	-6.95122
1.77E+12	0	-0.41895	-0.63965	0.623291	28.170732	50.731708	-10.3049
1.77E+12	0	-0.35889	-0.56396	0.605469	44.268295	29.695121	-12.561
1.77E+12	1	-0.07031	-0.2854	0.910156	45.853660	-4.20732	34.878047943115234
1.77E+12	0	-0.35718	-0.51636	0.615967	51.341464	25.182928	-7.13415
1.77E+12	1	-0.05078	-0.27783	0.911621	41.036586	-17.9268	36.09756088256836